

Virtual Robotics Testing & Optimization

Catch failures in simulation. Deploy robots faster, safer, cheaper.

Elevator pitch

“Our platform creates a virtual twin of your robot system and uses AI to systematically discover failures and optimize performance in simulation, so you can deploy robots faster, safer, and more efficiently.”

Value proposition

Physical testing of warehouse and service robots is slow, costly, and risky. Existing simulators (Gazebo, Webots, Isaac) provide physics — not an automated test lab. We add an intelligence layer: adversarial “what-if” search, domain randomization, and control optimization on a digital twin, so issues are found before hardware is at risk.

50%

Fewer physical test cycles
(estimate)

15%

Throughput gain from tuned
parameters

>95%

Target simulated mission success

6 mo

MVP: twin → adversarial engine

Who it is for

- **Warehousing:** fleets of pick/pack/carry robots — blocked aisles, sensor noise, speed vs safety.
- **Factory automation:** mobile manipulators — new arm/sensor layouts before you build them.
- **Service robots:** hotels and hospitals — emergency fallbacks (obstacle, low battery) in software first.

What makes it different

- Automatically generates adversarial stress tests instead of only running scripted scenes.
- Knowledge graph of robot, map, sensors, and tasks drives scenario search.
- Bayesian / RL optimization finds safe speed and gains; CBFs check safety margins.
- Human review is required — the platform never deploys changes on its own.

Ask / next step

Approve a six-month industrial-training MVP: Gazebo/ROS 2 digital twin (months 1–2), adversarial generator and Bayesian tuning (months 3–4), RL robustness and safety reports (months 5–6). Immediate proof: a 2D virtual lab that logs trajectories, detects collisions, and compares baseline vs blocked-aisle scenarios.

KPIs to track: failures found per sim-hour · coverage vs manual tests · throughput vs baseline · sim-to-real gap after demo.